



Ab initio many-body calculation of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ radiative capture

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ABSTRACT

We apply the ab initio no-core shell model/resonating group method (NCSM/RGM) approach to calculate the cross section of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ radiative capture. This reaction is important for understanding the solar neutrino flux. Starting from a selected similarity-transformed chiral nucleon–nucleon interaction that accurately describes two-nucleon data, we performed many-body calculations that simultaneously predict both the normalization and the shape of the S-factor. We study the dependence on the number of ${}^7\text{Be}$ eigenstates included in the coupled-channel equations and on the size of the harmonic oscillator basis used for the expansion of the eigenstates and of the localized parts of the integration kernels. Our S-factor result at zero energy is on the lower side of, but consistent with, the latest evaluation.

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The core temperature of the Sun can be determined with high accuracy through measurements of the ${}^8\text{B}$ neutrino flux, currently known with a $\sim 9\%$ precision [1]. An important input in modeling this flux is the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction [2] that constitutes the final step of the nucleosynthetic chain leading to ${}^8\text{B}$. At solar energies this reaction proceeds by external, predominantly nonresonant $E1$, S - and D -wave capture into the weakly-bound ground state (g.s.) of ${}^8\text{B}$. Experimental determinations of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ capture include direct measurements with proton beams on ${}^7\text{Be}$ targets [3–5] as well as indirect measurements through the breakup of a ${}^8\text{B}$ projectile into ${}^7\text{Be}$ and proton in the Coulomb field of a heavy target [6–8]. Theoretical calculations needed to extrapolate the measured S-factor to the astrophysically relevant Gamow energy were performed with several methods: the R-matrix parametrization [9], the potential model [10–12], microscopic cluster models [13–15] and, recently, also using the *ab initio* no-core shell model wave functions for the ${}^8\text{B}$ bound state [16]. The most recent evaluation of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S-factor at zero energy, $S_{17}(0)$, has a $\sim 10\%$ error dominated by the uncertainty in theory [2].

In this Letter, we present the first *ab initio* many-body calculations of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ capture starting from a nucleon–nucleon (NN) interaction that describes two-nucleon properties with high accuracy. We apply a recently developed technique that combines *ab initio* no-core shell model (NCSM) [17] and resonating-group method (RGM) [18,19] into a new many-body approach [20–22] (*ab initio* NCSM/RGM) capable of treating bound and scattering

states of light nuclei in a unified formalism. We use, in particular, the orthonormalized NCSM/RGM many-body wave functions given by

$$|\Psi^{J^\pi T}\rangle = \sum_{vv'} \int dr r^2 \int dr' r'^2 \hat{A}_v |\Phi_{vr}^{J^\pi T}\rangle \times \mathcal{N}_{vv'}^{-1/2}(r, r') \frac{\chi_{v'}^{J^\pi T}(r')}{r'}, \quad (1)$$

with the inter-cluster antisymmetrizer $\hat{A}_v$, the center-of-mass separation $\vec{r}_{A-a,a}$, and binary-cluster channel states

$$|\Phi_{vr}^{J^\pi T}\rangle = [(|A - a\alpha_1 I_1^{\pi_1} T_1\rangle |a\alpha_2 I_2^{\pi_2} T_2\rangle)^{(sT)} \times Y_\ell(\hat{r}_{A-a,a})]^{(J^\pi T)} \frac{\delta(r - r_{A-a,a})}{rr_{A-a,a}}. \quad (2)$$

The wave functions $\chi_v^{J^\pi T}(r)$ of the relative inter-cluster motion satisfy the integro-differential coupled-channel equations

$$\sum_{v'} \int dr' r'^2 [\mathcal{N}^{-\frac{1}{2}} \mathcal{H} \mathcal{N}^{-\frac{1}{2}}]_{vv'}(r, r') \frac{\chi_{v'}^{J^\pi T}(r')}{r'} = E \frac{\chi_v^{J^\pi T}(r)}{r} \quad (3)$$

with bound- or scattering-state boundary conditions. The Hamiltonian and norm kernels,

$$\mathcal{H}_{vv'}^{J^\pi T}(r', r) = \langle \Phi_{v'r'}^{J^\pi T} | \hat{A}_{v'} H \hat{A}_v | \Phi_{vr}^{J^\pi T} \rangle, \quad (4)$$

$$\mathcal{N}_{vv'}^{J^\pi T}(r', r) = \langle \Phi_{v'r'}^{J^\pi T} | \hat{A}_{v'} \hat{A}_v | \Phi_{vr}^{J^\pi T} \rangle, \quad (5)$$

contain all the nuclear structure and antisymmetrization properties of the problem. Further relevant details of the NCSM/RGM formalism are given in Ref. [20].

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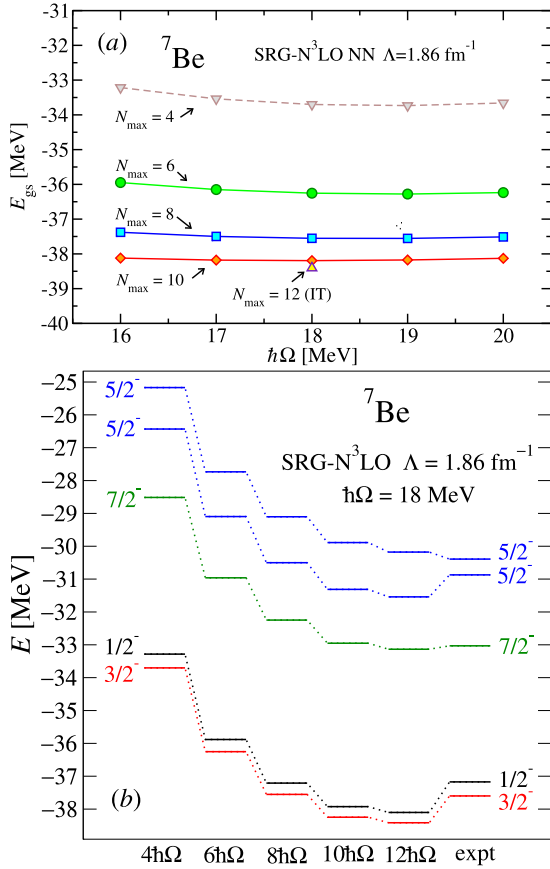


Fig. 1. (Color online.) Calculated NCSM ${}^7\text{Be}$ eigenenergies using the SRG- N^3LO NN potential with $\Lambda = 1.86 \text{ fm}^{-1}$. Panel (a) shows the dependence of the g.s. energy on the HO frequency for $N_{\text{max}} = 4$ –12. Absolute energies of the lowest 5 eigenstates for $N_{\text{max}} = 4$ –12 and $\hbar\Omega = 18 \text{ MeV}$ are compared to experimental values in panel (b). The $N_{\text{max}} = 12$ results were obtained within the importance-truncated basis.

In the present case A is equal to 8, and the projectile is a proton [$a = 1$ in Eq. (2)]. The input into Eq. (3) are: (i) the chiral N^3LO NN potential [23], which we soften by a similarity renormalization group (SRG) transformation [24,25] characterized by an evolution parameter Λ ; (ii) the eigenstates of the target, i.e. ${}^7\text{Be}$, calculated within the NCSM. In Fig. 1, we show the energy dependence of the ${}^7\text{Be}$ g.s. on the harmonic-oscillator (HO) frequency (a) for the HO basis sizes $N_{\text{max}} = 4$ to 12, with the 12 $\hbar\Omega$ results obtained using the importance-truncation scheme [26]. The frequency dependence is quite flat and, with the selected NN potential, we reach convergence for the g.s. at $N_{\text{max}} \approx 12$. The g.s. energy minimum is found at $\hbar\Omega = 18 \text{ MeV}$ and we choose this frequency for all subsequent calculations (including eigenstates and integration kernels). The convergence of the absolute energies of the lowest five ${}^7\text{Be}$ states is presented in panel (b) of Fig. 1. Compared to the experimental values, we observe a slight overbinding of the g.s. and an overestimation of the $7/2^-$ and $5/2^-$ state excitation energies, but, overall, the agreement is reasonable. In Table 1, we compare some of our (IT-)NCSM ${}^7\text{Be}$ results based on calculations up to $N_{\text{max}} = 14$ to experiment.

Table 1

${}^7\text{Be}$ g.s. energy (in MeV), charge radius (in fm), g.s. quadrupole (in $e \text{ fm}^2$) and magnetic (in μ_N) moments and M1 transition (in μ_N^2) obtained within the NCSM using the SRG- N^3LO NN potential with $\Lambda = 1.86 \text{ fm}^{-1}$. Experimental values are from Refs. [27,28].

	$E_{\text{g.s.}}$	r_c	Q	μ	$B(M1; \frac{1}{2}^- \rightarrow \frac{3}{2}^-)$
NCSM	−38.46	2.46(2)	−5.39(10)	−1.15	3.14
Expt.	−37.60	2.647(17)	–	−1.3995(5)	3.71(48)

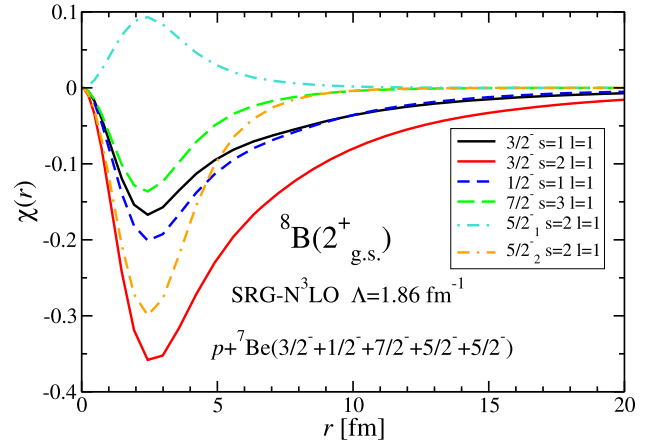


Fig. 2. (Color online.) Dominant P -wave components of the 2^+ ${}^8\text{B}$ g.s. wave function for $N_{\text{max}} = 10$ and $\hbar\Omega = 18 \text{ MeV}$, using the SRG- N^3LO NN potential with $\Lambda = 1.86 \text{ fm}^{-1}$. The NCSM/RGM calculation includes ${}^7\text{Be}$ g.s. and $1/2^-$, $7/2^-$, $5/2^-$ and $5/2^-$ excited states. The calculated s.e. is 136 keV.

Using the five lowest $N_{\text{max}} = 10$ eigenstates of ${}^7\text{Be}$, we first solve Eq. (3) with bound-state boundary conditions to find the g.s. of ${}^8\text{B}$. We note that the same N_{max} ($N_{\text{max}} + 1$ for the positive parity states) value is used to expand the localized parts of the integrations kernels (4) and (5). The chosen SRG- N^3LO NN potential with $\Lambda = 1.86 \text{ fm}^{-1}$ leads to a single bound state, 2^+ , with separation energy (s.e.) 136 keV, quite close to the experimental 137 keV. For the calculation of the low-energy behavior of the S_{17} S-factor, a correct s.e. is very important. The fact that the experimental s.e. of ${}^8\text{B}$ can be found using the SRG potential with a Λ from a “natural” range, i.e. ≈ 1.8 – 2.1 fm^{-1} , is reassuring. We also note that the three-nucleon (NNN) interaction induced by the SRG evolution of the NN potential is repulsive in this Λ range and, in very light nuclei, its contributions are cancelled to some extent by those of the initial attractive chiral NNN interaction (which is also SRG-evolved) [29–31]. Consequently, the present omission of both these NNN terms is better justified than it would be in other Λ ranges and/or in heavier nuclei. In Fig. 2, we plot the most significant components of the radial wave functions $\chi(r)$ for the 2^+ g.s. of ${}^8\text{B}$. The dominant component is clearly the channel-spin $s = 2$ P -wave of the ${}^7\text{Be}(\text{g.s.})$ - p that extends to a distance far beyond the plotted range. Remarkably, we notice a substantial contribution from the ${}^7\text{Be}(5/2^-)$ - p P -wave in the channel spin $s = 2$. (The other possible $s = 3$ P -wave configuration is negligible.) At the same time, the ${}^7\text{Be} 5/2^-$ state is dominated by a ${}^6\text{Li}$ - p channel-spin $s = 3/2$ P -wave configuration. Within the NCSM framework relevant to the present calculations this was shown (for the mirror ${}^7\text{Li}$ - n system) in Ref. [32]. Therefore, such a large contribution of the $s = 2$ ${}^7\text{Be}(5/2^-)$ - p P -wave to the ${}^8\text{B}$ ground state seems to indicate the presence of two antiparallel protons outside of a ${}^6\text{Li}$ core, and that their exchanges are important. Clearly, for a realistic description of the ${}^8\text{B}$ g.s., this state must be taken into account. The influence of still higher ${}^7\text{Be}$ resonances on the S-factor results will be discussed at the end of this Letter. From the ${}^8\text{B}$ g.s. wave functions, we can extract the asymptotic normalization coefficients (ANC). For example, for the ${}^7\text{Be}(\text{g.s.})$ - p components in the P -wave

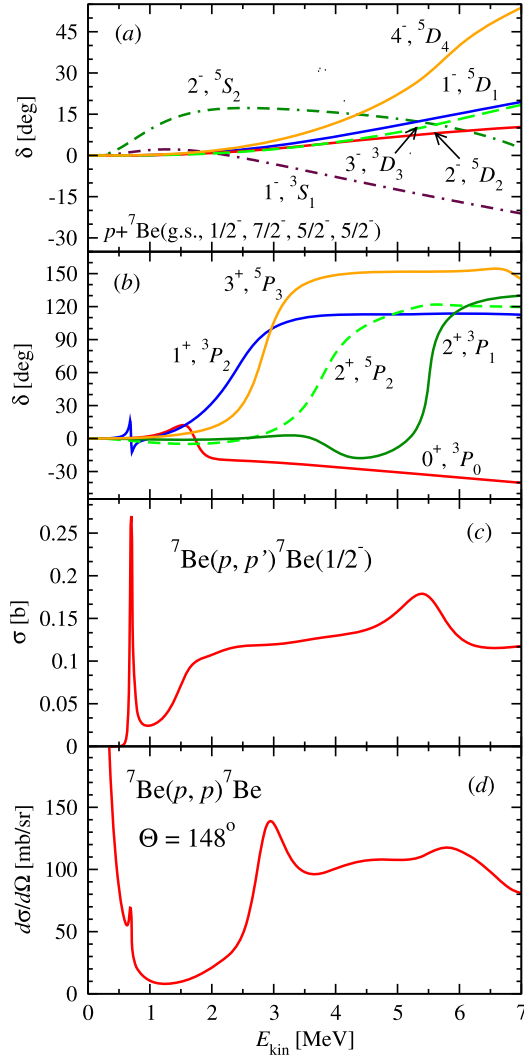


Fig. 3. (Color online.) Selected S- (a) D- (a), and P-wave (b) diagonal phase shifts of $p\text{-}^7\text{Be}$ elastic scattering, inelastic $^7\text{Be}(p, p')^7\text{Be}(1/2^-)$ cross section (c) and elastic $^7\text{Be}(p, p)^7\text{Be}$ differential cross section at $\Theta_{c.m.} = 148^\circ$ (d). Calculations as described in Fig. 2.

of relative motion and the channel spins $s = 1$ and $s = 2$ we obtained $C_{11} = 0.294 \text{ fm}^{-1/2}$ and $C_{12} = 0.650 \text{ fm}^{-1/2}$, respectively.

Next, we solve the same NCSM/RGM equations (3) with scattering-state boundary conditions for a chosen range of energies and obtain scattering wave functions and the scattering matrix. The resulting phase shifts and cross sections are displayed in Fig. 3. All energies are in the center of mass (c.m.). We find several P-wave resonances in the considered energy range. The first 1^+ resonance, manifested in both the elastic and inelastic cross sections, corresponds to the experimental ^8B 1^+ state at $E_x = 0.77 \text{ MeV}$ (0.63 MeV above the $p\text{-}^7\text{Be}$ threshold) [33]. The 3^+ resonance, responsible for the peak in the elastic cross section, corresponds to the experimental ^8B 3^+ state at $E_x = 2.32 \text{ MeV}$. However, we also find a low-lying 0^+ and additional 1^+ and 2^+ resonances that can be distinguished in the inelastic cross section. In particular, the $s = 1$ P-wave 2^+ resonance is clearly visible. There is also an $s = 2$ P-wave 2^+ resonance with some impact on the elastic cross section. These resonances are not included in the current $A = 8$ evaluation [33]. We note, however, that the authors of the recent Ref. [34] do claim observation of low-lying 0^+ and 2^+ resonances based on an R-matrix analysis of their $p\text{-}^7\text{Be}$ scattering experiment. Their

suggested 0^+ resonance at 1.9 MeV is quite close to the calculated 0^+ energy of the present work. While our calculated D-wave phase shifts show a slow monotonic increase, the S-wave phase shifts rise at low energies and then decrease, as seen in Fig. 3(a). This is a similar behavior as that found in microscopic three-cluster model calculations of Ref. [15], while in potential model calculations the S-wave phase shift monotonically decreases with energy [15]. Our extracted S-wave scattering lengths are -15.3 fm and -5.2 fm for the $s = 2$ and $s = 1$ channels, respectively. The experimental value for $s = 2$ scattering length is $-7(3) \text{ fm}$ [35]. The $s = 1$ scattering length has a very large uncertainty. We also note that our calculated scattering lengths decrease in absolute value with increasing values of the SRG parameter Λ : For $\Lambda = 2.02 \text{ fm}^{-1}$ we found an $s = 2$ S-wave scattering length of -10.2 fm [21]. Negative scattering lengths were also found in the cluster model of Ref. [15], although there the $s = 2$ scattering length was fitted to the experimental value mentioned above (-7 fm). The impact of the S-wave scattering length on the S-factor was also discussed in Refs. [12] and [36].

With the resulting bound- and scattering-state wave functions that are properly orthonormalized and antisymmetrized (1), we calculate the $^7\text{Be}(p, \gamma)^8\text{B}$ radiative capture using a one-body $E1$ transition operator. We use the one-body $E1$ operator defined in Eq. (3) of Ref. [37] that includes the leading effects of the meson-exchange currents through the Siegert's theorem. Here we note that any renormalization of the $E1$ operator brought about by the SRG procedure should be negligible as the $E1$ is a long-range operator while the SRG transformation is short range, when performed within the interval of Λ values used in the present calculations. The resulting S_{17} factor is compared to several experimental data sets in panel (a) of Fig. 4. In the data, one can see also the contribution from the 1^+ resonance due to the $M1$ capture that does not contribute to a theoretical calculation outside of the resonance and is negligible at astrophysical energies [2]. While the $M1$ operator poses more uncertainties than the Siegert's $E1$ operator (owing to the possible need of explicit two-body currents), its treatment within the NCSM/RGM formalism requires the evaluation of contributions from both the relative-motion part and the core (^7Be) part of the wave function and is only slightly more complicated compared to the $E1$ case. We plan to calculate its contribution in the future. Our calculated S-factor is somewhat lower than the recent Junghans data [5] but the shape reproduces closely the trend of the GSI data [8], which were extracted from Coulomb breakup. The shape is also quite similar to that obtained within the microscopic three-cluster model [15] (see the dashed line in Fig. 4(a)) used, after scaling to the data, in the most recent S_{17} evaluation [2]. The contributions from the initial 1^- , 2^- and 3^- partial waves are shown in panel (b) of Fig. 4. Our calculated $S_{17}(0) \approx 19.4 \text{ eV b}$ is on the lower side, but consistent with the latest evaluation $20.8 \pm 0.7(\text{expt}) \pm 1.4(\text{theory}) \text{ eV b}$ [2].

We studied the convergence of the ^7Be NCSM calculations in Fig. 1. To verify the behavior of our S-factor with respect to HO basis size and number of included ^7Be eigenstates, we performed additional calculations as summarized in Fig. 5. To study the dependence on the HO basis size, shown up to $N_{\text{max}} = 12$ in panel (b), we use the importance-truncation scheme and, due to computational limitations, we include only the three lowest eigenstates of ^7Be . The $N_{\text{max}} = 10$ and 12 S-factors are very close. In panel (a), we present results with up to eight ^7Be eigenstates obtained in an $N_{\text{max}} = 8$ basis. Calculations with more than five eigenstates are presently out of reach for larger N_{max} values. We can see a significant impact of the $5/2^-$ states (with only three ^7Be states, ^8B is unbound in this case). Among the others only the 8th state, $7/2^-$, contributes somewhat to the s.e. and flattens the S-factor at higher energies. We note that we selected different SRG- $N^3\text{LO}$ NN poten-

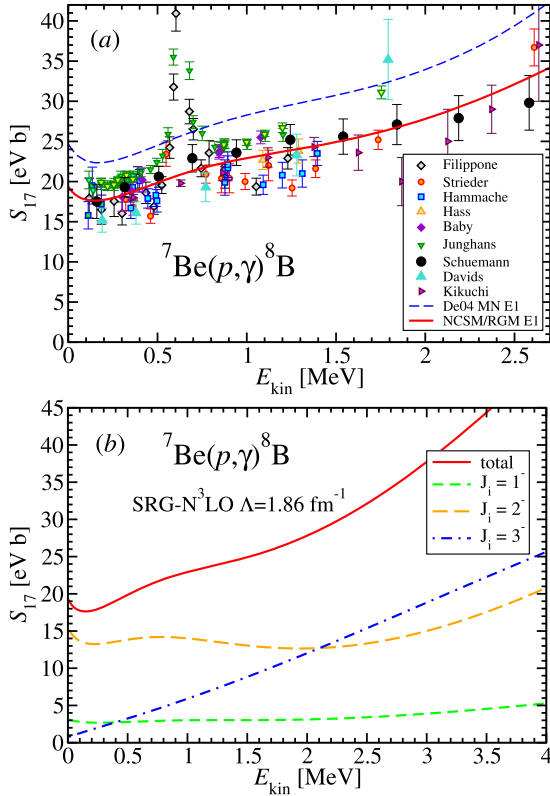


Fig. 4. (Color online.) Calculated ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S-factor as function of the energy in the c.m. compared to data and the microscopic cluster model calculations of Ref. [15] with the Minnesota (MN) interaction (a). Only E1 transitions were considered. Initial-state partial wave contributions are shown in panel (b). Calculation as in Fig. 2.

tials with the aim to match closely the experimental s.e. in each of the largest calculation. From these results we conclude that the use of the $N_{\text{max}} = 10$ space is justified and a limitation to the five lowest ${}^7\text{Be}$ eigenstates is quite reasonable (or that the $N_{\text{max}} = 8$ space is insufficient and a limitation to just 3 states is unrealistic). Also, based on these results we estimate the uncertainty of our calculated $S_{17}(0)$ to be ± 0.7 eV b.

An interesting feature of the S-factor is its flattening around 1.5 MeV. As seen in Fig. 4(b), this phenomenon is due to the S-wave contribution that dominates the $J_i = 2^-$ and 1^- partial waves at low energies. The increase of flattening with the number of ${}^7\text{Be}$ eigenstates included in the calculation, seen in Fig. 5(a), indicates that this is an effect due to the many-body correlations. This finding corroborates the observations of Ref. [15], where the flattening was attributed to the deformation of the ${}^7\text{Be}$ core. At the same time, we find that the flattening is somewhat correlated with the S-wave scattering length. For example, the S-wave $s = 2$ scattering length increases in absolute value from -13.4 fm, in the calculation with four states, to -14.5 fm, in that with the eight states [see Fig. 5(a)]. We also note that the flattening found in the present work is slightly larger than that obtained in the microscopic three-cluster model of Ref. [15], as seen in Fig. 4(a). Presumably, this is because in the three-cluster model the ${}^7\text{Be}$ structure was assumed to be of ${}^3\text{He}-{}^4\text{He}$ nature only, while the NCSM wave functions include in addition ${}^6\text{Li}-p$ configurations, particularly for the $5/2^-$ ${}^7\text{Be}$ state, as discussed earlier.

In Table 2, we summarize our calculated $S_{17}(0)$ S-factor, the S-wave scattering lengths and selected ANCs. We note that the ANCs from our *ab initio* approach are smaller than those obtained within the microscopic three-cluster model [15] (consistently with

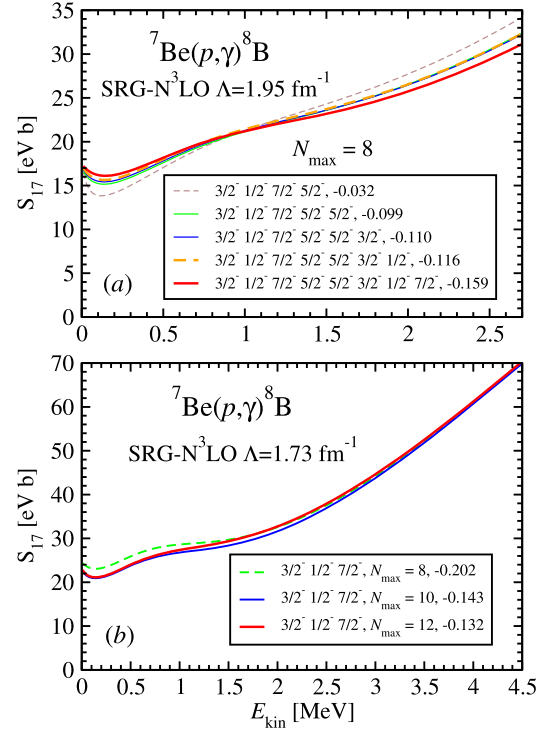


Fig. 5. (Color online.) Convergence of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S-factor with the number of ${}^7\text{Be}$ eigenstates (a) and the size of the HO basis used to expand the ${}^7\text{Be}$ eigenstates and localized parts of the integration kernels (b). The number of eigenstates and the calculated separation energy in each case is shown in the legend. HO frequencies of $\hbar\Omega = 19$ MeV (a) and 17 MeV (b) corresponding to the respective minima of ${}^7\text{Be}$ g.s. were used.

Table 2

Calculated $S_{17}(0)$ S-factor, the p - ${}^7\text{Be}$ S-wave scattering lengths, a_{01} ($s = 1$), a_{02} ($s = 2$) and the ${}^8\text{B}$ ground-state ANCs for ${}^7\text{Be}(\text{g.s.})-p$ in the P -wave and the channel spins $s = 1$ (C_{11}) and $s = 2$ (C_{12}). The NCSM/RGM calculation as described in Fig. 2. The $S_{17}(0)$ uncertainty was obtained as discussed in the text.

$S_{17}(0)$ [eV b]	a_{01} [fm]	a_{02} [fm]	C_{11} [$\text{fm}^{-1/2}$]	C_{12} [$\text{fm}^{-1/2}$]
19.4(7)	-5.2	-15.3	0.294	0.650

our smaller $S_{17}(0)$ value as seen from Fig. 4(a)) but in fairly good agreement with recent *ab initio* variational Monte Carlo calculations [38]. They are, however, still larger than the experimental ones from the DWBA analysis of Ref. [39].

The present calculations improve significantly the first NCSM-based ${}^7\text{Be}(p, \gamma){}^8\text{B}$ S-factor calculations [16] by a consistent treatment of both the bound and the scattering states. In Ref. [16], the scattering states were obtained within a potential model. The shape of the S-factor presented in this work clearly superseeds that found in Ref. [16]. The presently found $S_{17}(0)$ S-factor value is in between the values found in Ref. [16] with the two different NN potential models used there.

In conclusion, we performed *ab initio* many-body calculations of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ radiative capture that predict simultaneously both the normalization and the shape of the S-factor. Our S-factor result at zero energy, $S_{17}(0) = 19.4(7)$ eV b, is on the lower side of, but consistent with, the latest evaluation, and its shape follows closely the Coulomb breakup data from Ref. [8]. Our calculations can be further improved by including effects of additional higher-lying ${}^7\text{Be}$ resonances. This can be best done by coupling the NCSM/RGM binary-cluster basis with the NCSM calculations for ${}^8\text{B}$ as outlined in Ref. [40]. The inclusion of three-nucleon interactions, both chiral

and SRG-induced [29–31], is also desirable. Efforts in these directions are under way.

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